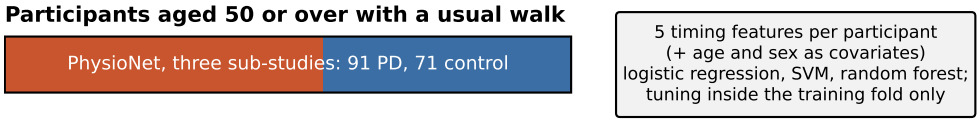
Participants aged 50 or over with a usual walk
PhysioNet, three sub-studies: 91 PD, 71 control
5 timing features per participant
(+ age and sex as covariates)
logistic regression, SVM, random forest;
tuning inside the training fold only
A. Within protocol: 5-fold cross-validation grouped by participant, repeated 20 times
train
train
train
train
test fold
Each fold holds about a fifth of the participants, stratified by group; a participant's recordings are never split between train and test. Reshuffled 20 times;
the mean over the 100 test folds is reported with a 95% interval corrected for the overlap between training sets.
B. Across protocols: leave one sub-study out
test: Ga (dual task), 47
Ju: 28 PD, 25 control
Si: 34 PD, 28 control
Ga: 29 PD, 18 control
test: Ju (cueing), 53
Si: 34 PD, 28 control
Ga: 29 PD, 18 control
Ju: 28 PD, 25 control
test: Si (treadmill study), 62
Train on two sub-studies, test on the third. The protocols, recording lengths and ages differ between sub-studies, so this is the primary comparison;
the interval is a bootstrap over the held-out participants.
C. Across cohorts: external test
train: all PhysioNet: 91 PD, 71 control
test: WearGait-PD, 100 PD, 85 control
A different research group and hardware; the same stride and swing timings from its pressure insoles or walkway. Run once, after the models are frozen.
WearGait-PD counts are from its article, before participants under 50 are removed.
PD
control
held out for testing